

MBS-010 - Future Directions Toward a Structural Science of Biological and Artificial Cognition

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Repository: <https://github.com/sizhet/MET-Brain-Structure-MBS>

Introduction

The MET Brain Structure (MBS) project is intentionally exploratory.

It does not claim to provide a complete theory of intelligence.

It does not claim to explain the brain.

It does not claim to solve cognition.

Instead, MBS proposes a structural perspective through which a variety of observations may be organized and investigated.

The true value of any structural framework ultimately depends on what it enables.

New questions.

New experiments.

New models.

New collaborations.

The purpose of this document is to outline possible future directions inspired by the MBS perspective.

These directions span neuroscience, cognitive science, artificial intelligence, autonomous systems, and human-AI collaboration.

MBS-001

Dream Runtime Simulator

One of the most natural extensions of MBS is the construction of a Dream Runtime Simulator.

The goal is not to reproduce biological dreaming.

Instead, the objective is to explore how:

- Trigger activation
- Retrieval systems

- Scene construction
- Functional tunnels

may interact within a simulated environment.

Potential questions include:

How do scenes emerge?

How do scenes transition?

How are functional pathways reinforced?

How might dream-like replay contribute to learning?

A simulator could provide a valuable experimental platform for investigating these questions.

MBS-002

Trigger Benchmark Framework

Current AI benchmarks largely focus on:

- Knowledge
- Reasoning
- Language
- Problem solving

MBS suggests another important dimension:

Triggering.

Future benchmark systems may evaluate:

- Significance detection
- Priority assignment
- Context sensitivity
- Attention allocation
- Trigger adaptation

Understanding triggering may prove as important as understanding reasoning.

A Trigger Benchmark Framework could become an important research tool.

MBS-003

Trigger Evolution Studies

The Trigger Convergence hypothesis raises many questions.

How do trigger categories emerge?

How do they change over time?

Which triggers remain stable?

Which triggers evolve rapidly?

Future research may investigate:

- Animal trigger systems
- Human trigger systems

- Social trigger systems
- Artificial trigger systems

Such studies could provide insight into the origins of cognitive convergence.

MBS-004

Cognitive Convergence Research

The MBS framework proposes:

Environment

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Triggers

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Retrieval

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Functional Tunnels

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Cognition

This hypothesis can be investigated empirically.

Future studies may examine:

- Cross-species convergence
- Human developmental convergence
- Cultural convergence
- Artificial agent convergence

The goal is not merely to observe convergence.

The goal is to identify structural mechanisms responsible for convergence.

MBS-005

Function Tunnel Discovery in Biological Systems

Function Tunnel Intelligence (FTI) inspired many aspects of MBS.

An important future question is:

Can biologically meaningful functional tunnels be identified?

Potential examples include:

Threat response tunnels.

Navigation tunnels.

Foraging tunnels.

Parenting tunnels.

Social interaction tunnels.

Future neuroscience research may reveal whether such recurring pathways possess identifiable structural signatures.

MBS-006

Retrieval Architecture Research

The Retrieval Layer occupies a central position within MBS.

Yet many questions remain.

How is retrieval organized?

How is prioritization achieved?

How do retrieval systems interact with triggering systems?

How do retrieval structures evolve?

Research in this area may benefit from collaboration among:

- Neuroscience
- Cognitive science
- Information retrieval
- Artificial intelligence
- Structural Intelligence

MBS-007

Embodied Scene Runtime Models

The Embodied Scene Runtime concept remains highly speculative.

However, it provides a promising direction for future investigation.

Potential research topics include:

- Scene activation
- Scene compression
- Scene transition
- Scene blending
- Scene prediction
- Scene-based planning

The long-term goal is to understand whether scenes represent a fundamental runtime unit of cognition.

MBS-008

Memory Beyond Storage

Traditional discussions often focus on memory as storage.

MBS encourages a broader perspective.

Future work may investigate memory as:

- Retrieval infrastructure
- Scene activation support
- Functional pathway reinforcement
- Trigger management

This shift may reveal new ways of understanding memory systems.

MBS-009

Human-AI Comparative Cognition

MBS naturally creates opportunities for comparison between biological and artificial systems.

Questions include:

What is a trigger in AI?

What is a scene in AI?

What is retrieval in AI?

What is a functional tunnel in AI?

What is cognitive convergence in AI?

Such comparisons may help identify both similarities and fundamental differences.

MBS-010

Autonomous Structural Intelligence

The relationship between MBS and Autonomous Structural Intelligence (ASI) deserves deeper exploration.

Both frameworks emphasize:

- Structural organization
- Runtime adaptation
- Continuous learning
- Functional pathways
- Trigger-driven behavior

Future work may investigate whether shared structural principles govern both biological and artificial cognition.

MBS-011

Human-AI Hybrid Cognition

The Human-AI Hybrid Civilization perspective introduces another possibility.

Future cognition may increasingly involve:

Human triggering.

AI retrieval.

Shared scene construction.

Collective functional tunnels.

Distributed adaptation.

Understanding these interactions may become one of the most important challenges of the coming decades.

MBS-012

Structural Neuroscience Collaboration

Perhaps the most important future direction involves collaboration.

MBS should not become isolated from neuroscience.

Its greatest value may emerge through dialogue with:

- Neuroscientists
- Cognitive scientists
- Psychologists
- Ethologists
- AI researchers
- Robotics researchers

The goal is not to defend a framework.

The goal is to learn from evidence.

Future discoveries should refine, modify, or replace parts of MBS whenever necessary.

MBS-013

Toward a Structural Science of Cognition

A longer-term possibility is the emergence of a broader structural science of cognition.

Such a science would seek common organizational principles across:

Brains.

Animals.

Humans.

AI systems.

Autonomous agents.

Hybrid intelligence systems.

The focus would not be specific implementations.

The focus would be structure.

How cognition organizes itself.

How cognition activates itself.

How cognition evolves itself.

How cognition converges.

The MBS project may represent one small step toward this broader goal.

A Research Roadmap

The future directions of MBS can be summarized as:

Dreaming

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Triggers

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Retrieval

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Scenes

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Function Tunnels

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Cognitive Convergence

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Biological Validation

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Artificial Intelligence

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Human-AI Hybrid Cognition

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Structural Science of Intelligence

Each stage presents opportunities for investigation.

Each stage presents opportunities for collaboration.

And each stage remains open to revision as new evidence emerges.

Final Reflection

The brain remains one of the greatest mysteries in science.

Artificial intelligence remains one of the most rapidly evolving technologies in history.

Understanding either system is difficult.

Understanding both simultaneously is even more difficult.

The MBS project does not claim certainty.

It offers a possibility.

A possibility that cognition may be organized through triggers, retrieval structures, scenes, tunnels, runtime activation, and continuous adaptation.

Whether this possibility ultimately proves useful will depend upon future research.

Yet even if parts of the framework change, the broader goal remains worthwhile:

To search for structural principles capable of connecting biological intelligence, artificial intelligence, and future forms of hybrid cognition.

That search is only beginning.